# Count groups of wiggles

Why Electronics Use Sine Waves

Name _____ Date _____

Show your thinking with numbers, marks, or words.

**1.** Little draws 2 groups of swing marks, with 3 marks in each group. How many marks are there?

3 + 3 = _____

Show your thinking.

**2.** Big draws spring bounces in 4 rows, with 2 bounces in each row. How many bounces are drawn?

2 + 2 + 2 + 2 = _____

Show your thinking.

**3.** Little counts 3 sets of rope wiggles, with 3 wiggles in each set. How many wiggles is that?

3 + 3 + 3 = _____

Show your thinking.

# Count groups of wiggles

Why Electronics Use Sine Waves

Name _____ Date _____

Show your thinking with numbers, marks, or words.

**4.** A page has 3 rows of wave drawings, with 4 drawings in each row. How many drawings are on the page?

$4 + 4 + 4 =$ _____

Show your thinking.

**5.** Big counts 5 groups of crank turns, with 3 turns in each group. How many turns does Big count?

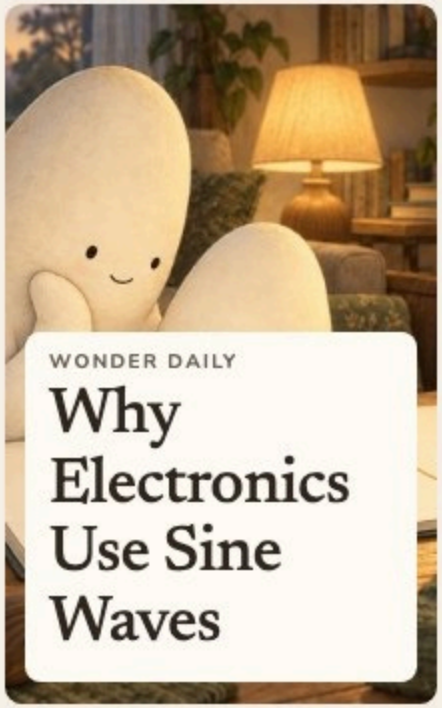
$3 + 3 + 3 + 3 + 3 =$ _____

Show your thinking.

**6.** Little draws speaker wiggles in 4 rows, with 5 wiggles in each row. How many wiggles are drawn?

$5 + 5 + 5 + 5 =$ _____

Show your thinking.

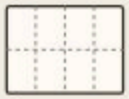


WONDER DAILY

# Why Electronics Use Sine Waves



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!





## A speaker's rhythm

The speaker's cone moved in and out to make a steady tone. Little drew its smooth rhythm. Where else might a wiggle appear?





## Turning into power

A generator turns motion into electrical energy. As a magnet spins beside wire, the electric push grows, fades, and changes direction smoothly.





## A bouncing spring

Big held a spring above a tray. Its end bounced up and down. Each bounce grew smaller until the spring settled.





## Slow at the turn

The swing slowed at each end and sped through the middle. A sine-like line could record its back-and-forth motion as time passed.





## A smooth wiggle

Little drew a smooth line that rose and fell again. Big called it a sine wave. Could this shape help explain a speaker?





## Along the rope

Little moved a rope from side to side. A wiggle traveled toward Big, while each bit of rope moved back and forth.

# Give a wiggle its words

Why Electronics Use Sine Waves · Curiosity day

Name _____ Date _____

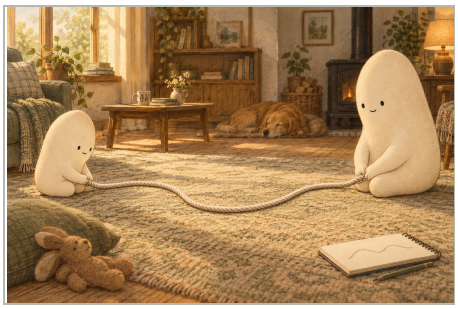
Say one clear command, then draw or write it.



Slow at the turn



A bounce comes back



A wiggle travels

Draw a swing, a spring, or a rope. Show which way it moves.

sine wave · wiggle · swing · spring · smooth · gently · back · forth

Swing … tell how to move gently

Watch … tell what to notice about a bounce

Draw … describe the smooth wiggle

Does your command tell what to do, using a word such as Swing, Watch, or Draw?